

Chemical constituents from the root bark of *Morus alba* and their chemotaxonomic significance

Xiao-Cong Liu, Xuan-Qin Chen, Rong-Tao Li, Zhi-Jun Zhang^{*}

Faculty of Life Science and Technology, Kunming University of Science and Technology, Kunming, 650500, Yunnan, PR China

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ABSTRACT

Research of detailed chemical constituents from the root bark of *Morus alba* resulted in the isolation of thirteen compounds, including a new triterpenoid (1), three lupane-type triterpenoids (2–4), two tirucallane-type triterpenoids (5, 6), three ursane-type triterpenoids (7–9), one incompletely cyclized triterpene alcohol (10), and three steroids (11–13). In this study, compound 1 was identified as a new triterpenoid. Among these known compounds, compounds 10 and 12 were firstly obtained from the Moraceae family, compound 5 was mentioned from the *Morus* genus for the first time, and this was the first report of compounds 6 and 13 in *Morus alba* species. Furthermore, chemotaxonomic significance of these isolates was also discussed in this paper.

1. Subject and source

Genus *Morus* is an important member of the Moraceae family which comprises around 15 species and 4 varieties around the world. *Morus alba* is a shrub which is widely distributed in Asia and mainly found in Japan, China, Korea and Vietnam. In China, *M. alba* is an economically significant plant, its leaves are indispensable food source used for rearing domesticated silkworm and its root bark has multiple uses in traditional Chinese medicine, which can be used for the treatment of cancer (Park and Park, 2022), neurological diseases (Chang et al., 2021), diabetes (Ha et al., 2018), inflammation (Vochýánová et al., 2017), depression (Lim et al., 2015), cough, lung-fire, hepatitis and heart diseases (Ha et al., 2016).

In the present study, the root bark of *M. alba* was collected from Chuxiong City, Yunnan Province, China (N25°48', E101°55'), and identified by one of the authors (Dr. Xuan-Qin Chen). A voucher specimen (number: KUST20210609) was deposited at the Faculty of Life Science and Technology, Kunming University of Science and Technology.

2. Previous work

Previous phytochemical investigations on the root bark of *M. alba* indicated the presence of various kinds of natural products, including stilbenes (Ha et al., 2016), isoprenylated flavonoids (Jung et al., 2015), coumarins (Kim et al., 2020), alkaloids and triterpenoids (Liu et al.,

2018). Among these compounds, stilbenes, isoprenylated flavonoids and coumarins have been extensively studied. However, studies identifying triterpenoids in the *M. alba* were scarce, and to the best of our knowledge, only a few triterpenoids have been previously isolated from the root bark of *M. alba* (Yang et al., 2011).

3. Present study

The isolation of EtOAc extracts of *M. alba* led to compounds 1–13, including a new triterpenoid (1), nine triterpenoids (2–10), and three steroids (11–13) (Fig. 1).

3.1. Extraction and isolation

The air-dried root bark of *M. alba* (14.8 kg) was cut into small pieces and extracted repeatedly with 75% EtOH (100 L × 3, each for two days) at room temperature. After removal of ethanol solution under reduced pressure, the residual was suspended in H₂O and partitioned sequentially with EtOAc and *n*-butanol. The EtOAc extract (1.3 kg) was subjected to silica gel column chromatography (200–300 mesh) eluted with step-gradient petroleum ether/EtOAc (100:1 to 5:1, v/v) to afford eight fractions (Fr.1–Fr.8). Fr.3 (33.6 g) was separated on reversed-phase column chromatography eluted with a gradient of MeOH/H₂O (40:60 to 100:0, v/v) to obtain five fractions (Fr.3-1–Fr.3-5), Fr.3-1 (1.2 g) was separated on Sephadex LH-20 with MeOH, and recrystallized with chloroform yield 3 (48.7 mg), 4 (32.6 mg) and 9 (18.5 mg). Fr.3-2 (5.1

^{*} Corresponding author.

E-mail address: zzj0121@126.com (Z.-J. Zhang).

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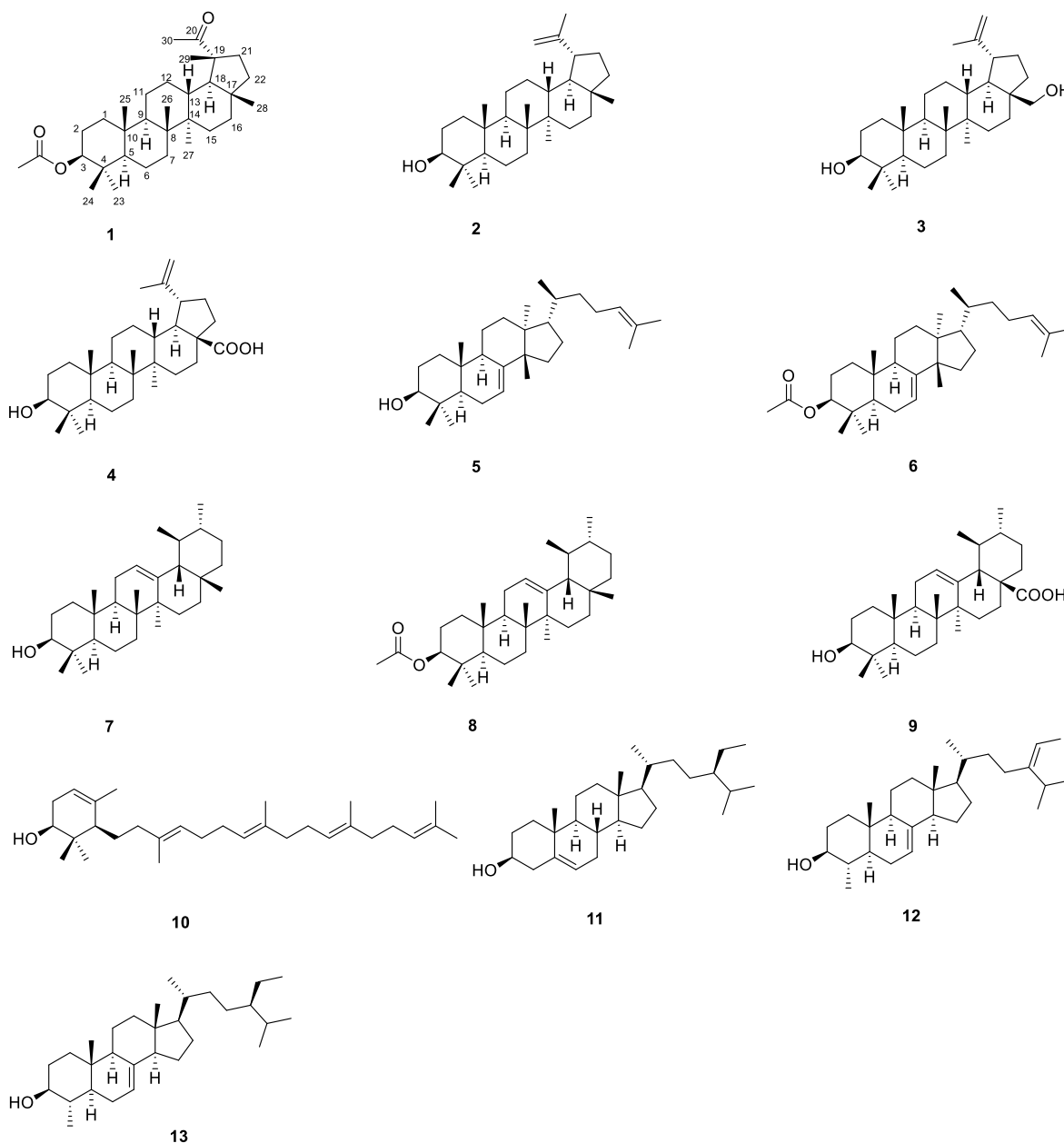


Fig. 1. Structures of compounds 1–13 isolated from *M. alba*.

g) was applied to a silica gel column (200–300 mesh) using a gradient solvent petroleum ether/EtOAc (8:1 to 2:1, v/v) to obtain two sub-fractions (Fr.3-2-1 and Fr.3-2-2), Fr.3-2-1 (3.1 g) and Fr.3-2-2 (2.0 g) were separately isolated on Sephadex LH-20 with MeOH to afford **1** (18.3 mg) and **10** (33.1 mg). Fr.3-3 (9.1 g) was separated on a silica gel column (200–300 mesh) eluted with step-gradient petroleum ether/EtOAc (15:1 to 2:1, v/v) to get six subfractions (Fr.3-3-1–Fr.3-3-6), Fr.3-3-3 (441.5 mg) was subjected to Sephadex LH-20 with MeOH, and further purified on preparative HPLC (85% MeOH in H₂O, 10 mL/min) to give **12** (7.7 mg), **13** (5.8 mg) and **11** (77.4 mg). Fr.3-3-4 (519.6 mg) was isolated on preparative HPLC (83% MeOH in H₂O, 10 mL/min) to produced **2** (11.3 mg), **5** (3.2 mg) and **7** (8.4 mg). Fr.3-4 (4.7 g) was isolated on a silica gel column (200–300 mesh) using a gradient solvent petroleum ether/EtOAc (45:1 to 5:1, v/v) to obtain five subfractions (Fr.3-4-1–Fr.3-4-5), Fr.3-4-2 (163.7 mg) was separated on Sephadex LH-20 with CHCl₃/MeOH (1:1, v/v), and then purified on preparative HPLC (90% MeOH in H₂O, 10 mL/min) to afford **6** (12.2 mg) and **8** (21.0 mg).

3.2. Structure elucidation

Compound **1** was isolated as a white amorphous powder. Its positive mode HR-ESIMS showed the pseudo-molecular ion peak $[M+Na]^+$ at m/z 507.3818 (calcd for 507.3809), which meant its molecular formula was C₃₂H₅₂O₃. The ¹H NMR spectrum (Table 1) showed the presence of nine singlet methyl groups [δ_H 0.83, 0.83, 0.84, 0.96, 0.98, 1.02, 1.20, 2.04, and 2.14], one methine proton attached with the acetoxy group [δ_H 4.45 (1H, dd, J = 11.0, 5.0 Hz, H-3)], and a methine proton [δ_H 1.95 (1H, d, J = 12.0 Hz, H-18)]. The ¹³C NMR and the DEPT spectra (Table 1) exhibited 32 signals, including nine methyl groups, ten methylene groups, five methines, and eight quaternary carbons. By comprehensive analysis of HRESIMS, ¹H NMR, ¹³C NMR, and 2D NMR of **1** showed similar patterns to those of 3 β -hydroxy-20-oxo-29(20 $\rightarrow$ 19)abeolupane (Chiang and Kuo, 2002), and the only difference was the presence of the other *O*-acetyl group [δ_H 2.04 (s); δ_C 171.1 (s) and 21.4 (q)] in **1**. The connection between the *O*-acetyl and C-3 (δ_C 80.9, d) was established by the HMBC correlations from the protons at δ_H 4.45 (1H,

Table 1¹H NMR (600 MHz) and ¹³C NMR (150 MHz) data of **1** in CDCl₃ (δ in ppm, J in Hz).

No.	δ _C	δ _H	No.	δ _C	δ _H
1	38.3	1.62 m, 0.96 m	17	43.4	
2	23.7	1.61 m	18	50.9	1.95 d (12.0)
3	80.9	4.46 dd (11.0, 5.0)	19	54.9	
4	37.8		20	214.1	
5	55.4	0.78 d (10.0)	21	37.8	1.88 m, 1.58 m
6	18.2	1.50 m, 1.38 m	22	40.6	1.58 m, 1.30 m
7	34.1	1.38 m	23	27.9	0.83 s
8	40.9		24	16.5	0.83 s
9	50.4	1.35 m	25	16.2	0.84 s
10	37.1		26	16.0	1.02 s
11	20.8	1.41 m, 1.35 m	27	15.2	0.98 s
12	25.3	1.21 m, 0.96 m	28	20.1	0.96 s
13	34.6	1.72 td (12.0, 2.0)	29	20.2	1.20 s
14	43.1		30	25.5	2.14 s
15	27.5	1.64 m, 1.02 m	OCOMe	171.1	
16	37.7	1.49 m, 1.44 m	OCOMe	21.4	2.04 s

dd, $J = 11.0, 5.0$ Hz, H-3) and 2.04 (s, -OCOMe) to the carbonyl group at δ_C 171.1 (s). The ROESY data showed that the relative configuration of **1** was the same as compound 3β-hydroxy-20-oxo-29(20 → 19) abeolupane (Fig. 2). The similar optical rotation of **1** and 3β-hydroxy-20-oxo-29(20 → 19) abeolupane ($[\alpha]_D^{25} = +23.5$ (c 0.3, MeOH), 3β-hydroxy-20-oxo-29(20 → 19) abeolupane: $[\alpha]_D^{21} = +16.3$ (c 0.2, CHCl₃)) (Chiang and Kuo, 2002) indicated that **1** has the same absolute configuration as 3β-hydroxy-20-oxo-29(20 → 19) abeolupane. Therefore, the structure of **1** was elucidated as 3β-O-acetyl-20-oxo-29(20 → 19) abeolupane.

Compounds **2–13** were identified by a combination of spectroscopic methods (¹H, ¹³C NMR and DEPT) and comparisons with the literature data as lupeol (**2**) (Im et al., 2006), betuline (**3**) (Wang et al., 2017), betulinic acid (**4**) (Wang et al., 2017), butyrospermol (**5**) (Zare et al., 2015), butyrospermol acetate (**6**) (Filip et al., 2015), α-amyrin (**7**) (Saeed and Sabir., 2003), α-amyrin acetate (**8**) (Lima et al., 2004), ursolic acid (**9**) (Alves et al., 2000), camelliols C (**10**) (Akihisa et al., 1999a,b), β-sitosterol (**11**) (Lee et al., 2013), sitrostadienol (**12**) (Yan et al., 2015), 24-ethyl-lophenol (**13**) (Yan et al., 2015).

4. Chemotaxonomic importance

In this paper, we report the isolation and structure elucidation of a new triterpenoid (**1**), together with twelve known compounds including three lupane-type triterpenoids (**2–4**), two tirucallane-type triterpenoids (**5, 6**), three ursane-type triterpenoids (**7–9**), one incompletely cyclized triterpene alcohol (**10**), and three steroids (**11–13**).

Compound **1** was a new one and named as 3β-O-acetyl-20-oxo-29(20

→ 19) abeolupane, and its similar 3β-hydroxy-20-oxo-29(20 → 19) abeolupane was only obtained from *Ficus macrocarpa*. This is the first report of the presence of compound **1** from the genus *Morus* as well as demonstrate some relationships between *Morus* and *Ficus*. Therefore, we think that compound **1** could serve as chemotaxonomic markers of genus *Morus*.

Here, we reported three lupane-type triterpenoids (**2–4**), Compound **2** was previously isolated from several plants, such as *Lippia rugosa* (Nono et al., 2022), *Maesa lanceolata* (Seumo et al., 2022), *Moringa oleifera* (Vasanth et al., 2022), *Morus macroura* (Hamdan et al., 2022), *Desmodium salicifolium* (Donkia et al., 2022). Compound **3** was previously reported in *Alstonia scholaris* (Sultana et al., 2020), *Machilus salicina* (Yang et al., 2022), *Helicteres hirsuta* (Tra et al., 2021), *Sophora mollis* (Quradha et al., 2021), *Celastrus orbiculatus* (Vu et al., 2021), *Diospyros gillettii* (Jouwa Tameye et al., 2020) and many other plants. Compound **4** was previously identified from several kinds of plants, including *Burkea africana* (Feunaing et al., 2022), *Tabernaemontana inconspicua* (Nghah et al., 2022), *Desmodium salicifolium* (Donkia et al., 2022), *Syzygium aromaticum* (Kiran et al., 2022), *Combretum quadrangulare* (Nguyen et al., 2022), *Senna siamea* (Chedjou et al., 2022), *Combretum glutinosum* (Toklo et al., 2021). To sum up, compounds **2–4** were broadly distributed in a variety of plants including the genus *Morus*. So, they are not very characteristic components from genus *Morus*.

According to the literature, compounds **5** and **6** were identified as tirucallane-type triterpenoids. Compound **5** was isolated from the *Morus* genus for the first time. As the same time, Compound **5** was previously obtained from *Clusia burle-marxii* (Ribeiro et al., 2019), and *Euphorbia fischeriana* (Yu and Hou, 2010). Thus, compound **5** may be a useful taxonomic marker for genus *Morus*. Besides, compound **6** was found in *Maclura pomifera* (Filip et al., 2015), *Morus macroura* (Wu et al., 2003). Above the two plants belong to the Moraceae family, which could provide more chemotaxonomic evidence for this family.

Three ursane-type triterpenoids **7–9** were broadly distributed in several plants including the *Eucalyptus globulus* (Ibrahim et al., 2013), *Trachelospermum jasminoids* (Zhang et al., 2012), *Ficus pandurata* (Ramadan et al., 2009) and so on. From above mentioned, compounds **7–9** are not considered as valuable markers to support the taxonomic classification of *M. alba* in that they were common in many plants.

One incompletely cyclized triterpene alcohol **10** was previously obtained from *Quercus robur* (Unuofin and Lebelo, 2021), *Euphorbia antiquorum* (Akihisa et al., 2002), *Garcinia speciosa* (Rukachaisirikul et al., 2003), *Camellia sasanqua* (Akihisa et al., 1999a,b), and was first time isolated from family Moraceae. Obviously, compound **10** provides new information on the chemistry for *M. alba* and may serve as valued chemotaxonomic markers for *M. alba*.

Among three steroids (**11–13**), compound **12** was previously gave from *Ziziphus jujuba* (Elaloui et al., 2016), *Vaccinium myrtillus* (Szakiel

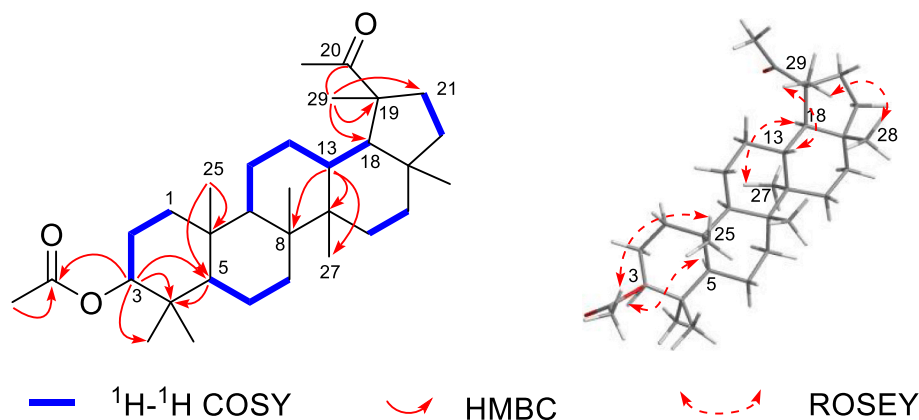


Fig. 2. Key ¹H-¹H COSY, HMBC, ROSEY correlations of compound **1**.

et al., 2012), *Linum usitatissimum* (Herchi et al., 2009), *Cordyceps militaris* (Lv et al., 2008). This was the first report on compound **12** isolated from the family Moraceae. Therefore, compound **12** could be considered as the specific markers of *M. alba*. Compound **13** was previously obtained from *Prunus mume* (Yan et al., 2015), *Olive* (Stiti et al., 2007), *Mulberry cultivars* (Zambakhidze et al., 2005), *Vanilla madagascariensis* (Ramaroson-Raonizafinimanana et al., 1998). The findings verified that these genera may share similar biosynthesis pathways for metabolites.

In summary, thirteen compounds were isolated from the roots bark of *Morus alba*. More importantly, compound **1** was identified as a new triterpenoid, butyrospermol (**5**) were first reported from the genus *Morus*, butyrospermol acetate (**6**) and 24-ethyl-lophenol (**13**) were first isolated from species *Morus alba*, and camelliols C (**11**) and sitrostadienol (**12**), were first obtained from the family Moraceae. Thus, compounds **1**, **5**, **6**, **11**, and **13** may serve as potential chemotaxonomic markers for *M. alba* and could be used to distinguish among species of Moraceae.

CRedit authorship contribution statement

Xiao-Cong Liu: Writing-original draft, Investigation. **Xuan-Qin Chen:** Resources. **Rong-Tao Li:** Project administration, Funding acquisition. **Zhi-Jun Zhang:** Formal analysis, Project administration, Funding acquisition, Conceptualization, Methodology, Supervision, Writing-review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.bse.2023.104585>.

References

- Akihisa, T., Arai, K., Kimura, Y., Koike, K., Kokke, W.C.M.C., Shibata, T., Nikaido, T., 1999a. *J. Nat. Prod.* 62, 265–268.
- Akihisa, T., Kithsirij Wijeratne, E.M., Tokuda, H., Enjo, F., Toriumi, M., Kimura, Y., Koike, K., Nikaido, T., Tezuka, Y., Nishino, H., 2002. *J. Nat. Prod.* 65, 158–162.
- Akihisa, T., Koike, K., Kimura, Y., Sashida, N., Matsumoto, T., Ukiya, M., Nikaido, T., 1999b. *Lipids* 34, 1151–1157.
- Alves, J.S., Castro, A.J.C.M., Freire, M.O., Cunha, E.V.L., Barbosa-Filho, J.M., Silva, M.S., 2000. *Magn. Reson. Chem.* 38, 201–206.
- Chang, Y., Hung, C.F., Ko, H.H., Wang, S.J., 2021. *J. Med. Food* 24, 209–217.
- Chedjou, I.N., Ngouafong, F.T., Tchuengue, R.T., Dzoyem, J.P., Ponou, B.K., Teponno, R.B., Barboni, L., Tapondjou, L.A., 2022. *Nat. Prod. Res.* 10, 1–9.
- Chiang, Y.M., Kuo, Y.H., 2002. *J. Org. Chem.* 67, 7656–7661.
- Donkja, A.P.T., Eckhardt, P., Tsafack, B.T., Tchuengue, R.T., Groß, J., Ponou, B.K., Teponno, R.B., Dzoyem, J.P., Opatz, T., Barboni, L., Tapondjou, L.A., 2022. *Phytochem. Lett.* 50, 100–105.
- Elaloui, M., Laamouri, A., Ennajah, A., Cerny, M., Mathieu, C., Vilarem, G., Chaar, H., Hasnaoui, B., 2016. *Ind. Crop. Prod.* 83, 133–139.
- Feunaing, R.T., Tamfu, A.N., Ntchapda, F., Gade, I.S., Mbane, M.N., Tagatsing, M.F., Talla, E., Henoumont, C., Laurent, S., Dinica, R.M., 2022. *Nat. Prod. Res.* 36, 4132–4139.
- Filip, S., Djarmati, Z., Lisichkov, K., Csanadi, J., Jankov, R.M., 2015. *Ind. Crop. Prod.* 76, 995–1000.
- Ha, M.T., Seong, S.H., Nguyen, T.D., Cho, W.K., Ah, K.J., Ma, J.Y., Woo, M.H., Choi, J.S., Min, B.S., 2018. *Phytochemistry* 155, 114–125.
- Ha, M.T., Tran, M.H., Ah, K.J.K., Jo, K.J., Kim, J., Kim, W.D., Cheon, W.J., Woo, M.H., Ryu, S.H., Min, B.S., 2016. *Bioorg. Med. Chem. Lett.* 26, 2788–2794.
- Hamdan, D.I., Hafez, S.S., Hassan, W.H.B., Morsi, M.M., Khalil, H.M.A., Ahmed, Y.H., Ahmed-Farid, O.A., El-Shiekh, R.A., 2022. *RSC Adv.* 12, 3476–3493.
- Herchi, W., Harrabi, S., Sebei, K., Rochut, S., Boukhchina, S., Pepe, C., Kallel, H., 2009. *Plant Physiol. Biochem.* 47, 880–885.
- Ibrahim, M., Ambreen, S., Hussain, A., Hussain, N., Imran, M., Ali, B., Sumrra, S.H., Yousof, M., Rehmani, F.S., 2013. *Asian J. Chem.* 26, 1011–1014.
- Im, K.R., Jeong, T.S., Kwon, B.M., Baek, N.I., Kim, S.H., Kim, D.K., 2006. *Arch. Pharm. Res. (Seoul)* 29, 191–194.
- Jouwa-Tameye, N.S., Akak, C.M., Happi, G.M., Frese, M., Stammler, H.G., Neumann, B., Lenta, B.N., Sewald, N., Nkengfack, A.E., 2020. *Phytochem. Lett.* 36, 63–67.
- Jung, J.W., Ko, W.M., Park, J.H., Seo, K.H., Oh, E.J., Lee, D.Y., Lee, D.S., Kim, Y.C., Lim, D.W., Han, D., Baek, N.I., 2015. *Arch. Pharm. Res. (Seoul)* 38, 2066–2075.
- Kim, J.H., Doh, E.J., Lee, G., 2020. *Molecules* 25, 25235592.
- Kiran, Z., Khan, H.N., Rasheed, S., Begum, S., Iqbal Choudhary, M., Sara, Bano Z., Siddiqui, B.S., Fayyaz, S., Iqbal, E., Hussain, T., Lateef, M., Atta Ur, R., 2022. *Nat. Prod. Res.* 23, 1–6.
- Lee, J.M., Lee, D.G., Lee, K.H., Cho, S.H., Nam, K.W., Lee, S., 2013. *Afr. J. Pharm. Pharmacol.* 7, 294–301.
- Lim, D.W., Jung, J.W., Park, J.H., Baek, N.I., Kim, Y.T., Kim, I.H., Han, D., 2015. *Biol. Pharmaceut. Bull.* 38, 1772–1778.
- Lima, M.P., Braga, P.A.C., Macedo, M.L., Silva, M.F.G.F., Ferreira, A.G., Fernandes, J.B., Vieira, P.C., 2004. *J. Braz. Chem. Soc.* 15, 385–394.
- Liu, B.R., Yan, T.N., Xiao, J., Wang, X.L., 2018. *Chinese Herbal Medicines*, vol. 10, pp. 331–335.
- Lv, Z.M., Jiang, Y.T., Wu, L.J., Liu, K., 2008. *China J. Chin. Mater. Med.* 33, 2914–2917.
- Ngah, L., Tsopgni, W.D.T., Nyobe, J.C.N., Tcho, A.T., Langat, M.K., Ndom, J.C., Mas-Claret, E., Sadgrove, N.J., Waffo, A.F.K., Phumthum, M., 2022. *Antibiotics-Basel* 11, 11010121.
- Nguyen, H.H., Bui, X.H., Tran, T.M., Sichaem, J., Le, T.K., Nguyen, N.H., Duong, T.H., 2022. *J. Asian Nat. Prod. Res.* 24, 691–696.
- Nono Nono, E.C., Tsopmejo, J.P., Momeni, J., Nkouam, T.F., Abdou, J.P., Choudhary, M. I., Nkengfack, A.E., 2022. *Nat. Prod. Res.* 15, 1–9.
- Park, H.J., Park, S.H., 2022. *Nutr. Cancer. An Int. J.* 74, 1837–1848.
- Quradha, M.M., Khan, R., Adhikari, A., Rauf, A., Rashid, U., Bawazeer, S., Al-Awthan, S. Y., Bahattab, O., Mubarak, M.S., 2021. *ACS Omega* 6, 15911–15919.
- Ramadan, M.A., Ahmad, A.S., Nafady, A.M., Mansour, A.I., 2009. *Nat. Prod. Res.* 23, 1218–1230.
- Ramaroson Raonizafinimanana, B., Gaydou, E.M., Bombarda, I., 1998. *J. Am. Oil Chem. Soc.* 75, 1325–1328.
- Ribeiro, P.R., Ferraz, C.G., Cruz, F.G., 2019. *Biochem. Systemat. Ecol.* 82, 31–34.
- Rukachaisirikul, V., Pailee, P., Hiranrat, A., Tuchinda, P., Yoosook, C., Kasitsit, J., Taylor, W.C., Reutrakul, V., 2003. *Planta Med.* 69, 1141–1146.
- Saeed, M.A., Sabir, A.W., 2003. *J. Asian Nat. Prod. Res.* 5, 35–41.
- Seumo, A.S., Nanfack, A.R.D., Ndonsa, B.L., Bitchagno, G.T.M., Mbouangouere, R., Lenta, B.N., Sewald, N., Tane, P., Tane, M., Ngouela, S.A., 2022. *Nat. Prod. Res.* 36, 4379–4387.
- Stiti, N., Triki, S., Hartmann, M.A., 2007. *Lipids* 42, 55–67.
- Sultana, N., Qazi, M.S., Kamal, M., 2020. *Anti-Inflammatory and Anti-allergy Agents in Medicinal Chemistry*, vol. 19, pp. 370–386.
- Szakiel, A., Pączkowski, C., Huttunen, S., 2012. *J. Agric. Food Chem.* 60, 11839–11849.
- Toklo, P.M., Yayi Ladekan, E., Linden, A., Hounzangbe-Adote, S., Kouam, S.F., Gbenou, J.D., 2021. *Acta Crystallogr. C. Struct. Chem.* 77, 505–512.
- Tra, N.T., Cham, B.T., Ha, N.T.T., Anh, L.T.T., Tuyen, N.V., Son, N.T., 2021. *Nat. Prod. J.* 11, 592–599.
- Unuofin, J.O., Lebelo, S.L., 2021. *Arab. J. Chem.* 14, 102970.
- Vasanth, K., Minakshi, G.C., Velu, K., Priya, T., Mohan-Kumar, R., Kaliappan, I., Dubey, G.P., 2022. *J. Food Biochem.* 46, e14170.
- Vočyánová, Z., Pokorná, M., Rotrekl, D., Šmekal, V., Fictum, P., Suchý, P., Gajdziok, J., Šmejkal, K., Hošek, J., 2017. *PLoS One* 12, e0182464.
- Vu, T.O., Tran, P.T., Seo, W., Lee, J.H., Min, B.S., Kim, J.A., 2021. *J. Nat. Med.* 75, 56–65.
- Wang, C.M., Tsai, S.J., Jhan, Y.L., Chou, C.H., 2017. *Molecules* 22, 22122119.
- Wu, F., Sun, S.G., Chen, R.Y., 2003. *China J. Chin. Mater. Med.* 28, 141–143.
- Yan, X.T., Lee, S.H., Li, W., Jang, H.D., Kim, Y.H., 2015. *Arch. Pharm. Res. (Seoul)* 38, 186–192.
- Yang, S.Y., Jiang, G.H., Sun, Q.H., Luo, Q., Wang, C.X., Liu, B., Zhan, R., Aisa, H.A., Chen, Y.G., 2022. *Biochem. Systemat. Ecol.* 103, 104432.
- Yang, Z.G., Matsuzaki, K., Takamatsu, S., Kitanaka, S., 2011. *Molecules* 16, 6010–6022.
- Yu, M., Hou, A., 2010. *China Journal of Chinese Materia Medica*, pp. 3002–3006.
- Zambakhidze, N.E., Sulaberindze, K.V., Mzhavanadze, V.V., Tsiklauri, G., 2005. *Appl. Biochem. Microbiol.* 41, 460–462.
- Zare, S., Ghaedi, M., Miri, R., Heiling, S., Asadollahi, M., Baldwin, I.T., Jassbi, A.R., 2015. *Iran. J. Pharm. Res. (IJPR)* 14, 243–249.
- Zhang, J., Yin, Z.Q., Liang, J.Y., Diao, Y.J., 2012. *Chin. Tradit. Patent Med.* 34, 1939–1942.